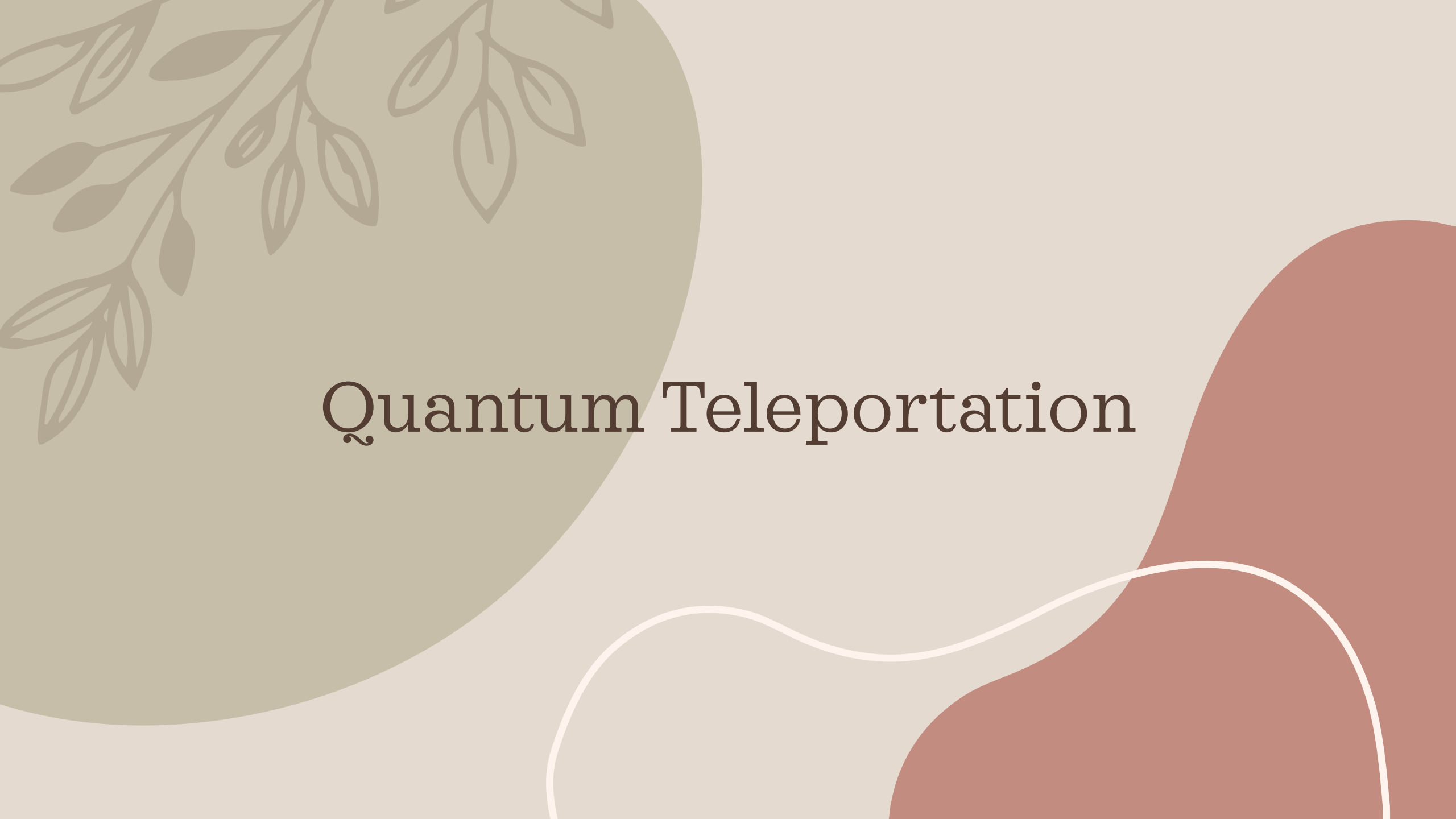




Quantum Teleportation

Teleportation & Quantum Communication

- Teleportation here refers to transferring information from one location to another (not copying)
- Since the states of qubits cannot be copied (non-cloning theorem), teleportation is the main approach for passing the state of one qubit to another qubit on the network
- In what follows we describe how teleportation is used to solve this problem

Quantum Teleportation

- Quantum teleportation, is a protocol where a sender (Alice) transmits a qubit to a receiver (Bob) by making use of a shared entangled quantum state along with two bits of “**classical communication**”.
- The word *teleportation* usually refers to transportation of solid matter from one place to another using some futuristic communication process, but is used here to refer to information transportation — what is actually teleported is quantum information.

What is Information Teleportation?

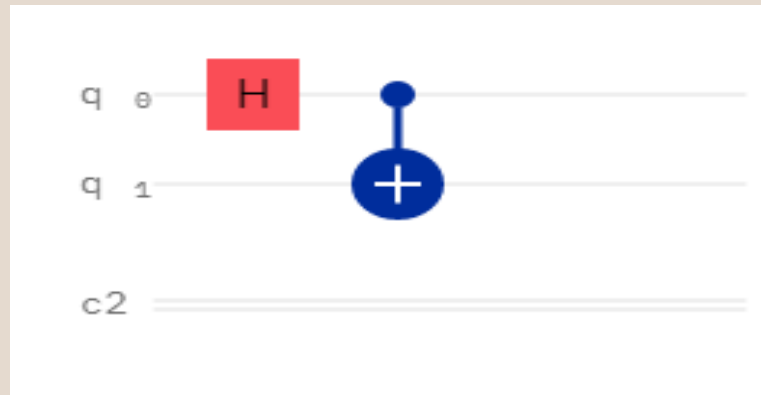
- Quantum teleportation is a technique to transfer a quantum state

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

from one qubit to another, often separated by a large distance.

- This process is based on quantum entanglement and classical communication to achieve state transfer without physically moving the qubit.

Teleportation is Strongly Based on Entanglement



The Setup Process

Teleportation is preceded by a setup process that may be described as follows:

- It is assumed that Alice and Bob share an entangled state.
- Alice holds a qubit **A**, Bob holds a qubit **B**, and together the pair **(A, B)** is in the state $|\phi^+\rangle$.
- It could be, for instance, that Alice and Bob were in the same location in the past, they prepared the qubits **A** and **B** in the state $|\phi^+\rangle$, and then each went their own way with their qubit in hand.
- This implies that distance has no effect on the entangled state. Here each of Alice and Bob took his/her qubit away, and perhaps traveled to a different part of the world
- It may also be the case that the setup of the two qubits has been done in a lab or factory, then distributed to the respective parties.

The Teleportation Protocol

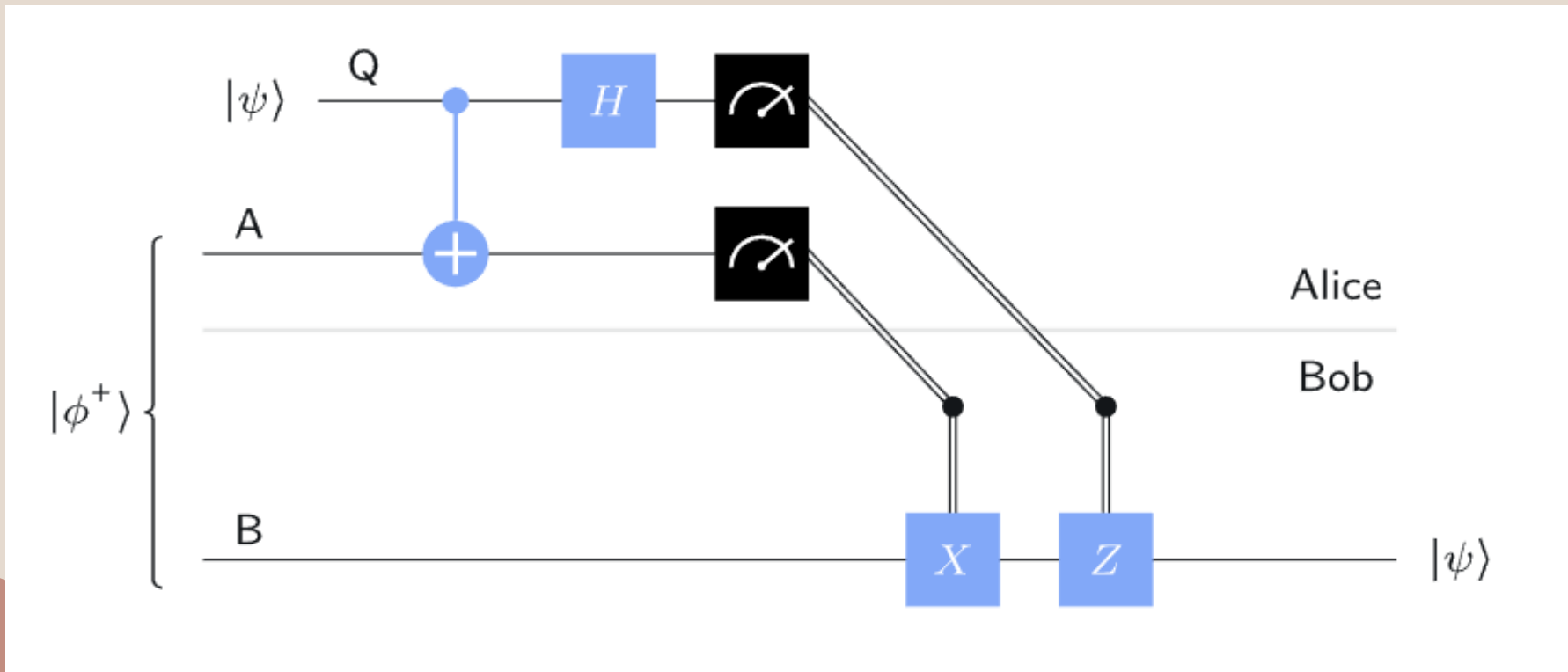


Diagram Details

- In the above diagram, the Q qubit represents that the state that should be sent to Bob
- A is Alice's qubit, while B is Bob's qubit, Already entangled, but are now in different locations

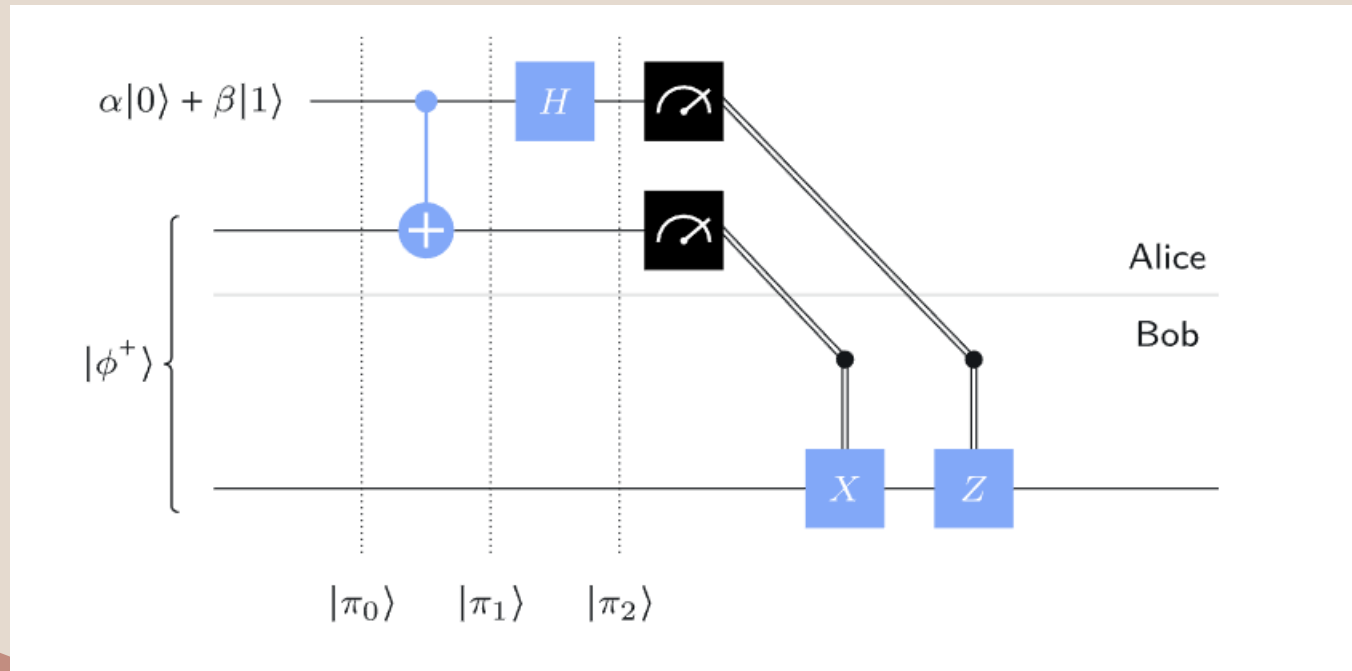
The Protocol

The teleportation protocol may be described as follows:

1. Alice performs a controlled-NOT operation on the pair (A, Q) , with Q being the control and A being the target, and then performs a Hadamard operation on Q .
2. Alice then measures both A and Q , with respect to a standard basis measurement in both cases, and transmits the classical outcomes to Bob. Let us refer to the outcome of the measurement of A as a and the outcome of the measurement of Q as b .
3. Bob receives a and b from Alice, and depending on the values of these bits he performs these operations:
 1. If $a = 1$, then Bob performs a bit flip (or X gate) on his qubit B .
 2. If $b = 1$, then Bob performs a phase flip (or Z gate) on his qubit B .
4. That is, conditioned on ab being 00, 01, 10, or 11, Bob performs one of the operations I , Z , X , or ZX on the qubit B .

Applying the protocol : A simple Example

The following diagram shows a simple example that applies the protocol



Applying the protocol

- Under the assumption that the qubit **Q** begins the protocol in the state $\alpha | 0 \rangle + \beta | 1 \rangle$, the state of the three qubits (**B**, **A**, **Q**) together at the start of the protocol is therefore: result at position π_0

$$|\pi_0\rangle = |\phi^+\rangle \otimes (\alpha|0\rangle + \beta|1\rangle) = \frac{\alpha|000\rangle + \alpha|110\rangle + \beta|001\rangle + \beta|111\rangle}{\sqrt{2}}.$$

- The first gate that is performed is the controlled-NOT gate, which transforms the state $|\pi_0\rangle$ into the expression at π_1 :

$$|\pi_1\rangle = \frac{\alpha|000\rangle + \alpha|110\rangle + \beta|011\rangle + \beta|101\rangle}{\sqrt{2}}.$$

- Then the Hadamard gate is applied, which transforms the state $|\pi_1\rangle$ into: (position π_2)

$$|\pi_2\rangle = \frac{\alpha|00\rangle|+\rangle + \alpha|11\rangle|+\rangle + \beta|01\rangle|-\rangle + \beta|10\rangle|-\rangle}{\sqrt{2}}$$

Applying the Protocol

- Proceeding with above analysis, it will be realized the outcome of the (BAQ) combination is:

$$(\alpha|1\rangle - \beta|0\rangle)|11\rangle.$$

Now Bob Applies ZX on the qubit B to get:

$$(\alpha|0\rangle + \beta|1\rangle)|11\rangle.$$